

Niccolò **Laurenti**

PH.D. IN THEORETICAL PARTICLE PHYSICS · SOFTWARE DEVELOPER

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Experience

Software Test Solutions Engineer

Rome, Italy

AT TMC ITALY — *Jan. 2026 - Current*

- Worked for **TMC Italy** as a consultant for **ELT Group** in the Test Solutions department.
- Developed and automatized software components test procedures using **Robot Framework** test suites, integrating **Python** and **C++** simulators, to improve software reliability.
- Performed system-level integration and Verification & Validation activities to verify system functionality and ensure compliance with technical requirements.

Technologies: Robot Framework, Python, C++, Qt, CMake, Conan, Git, Gitlab, Docker, Bash, JSON, YAML, VS Code, Microsoft Copilot, Matlab, Red Hat, Microsoft Office, Microsoft Teams

Embedded Software Developer

Rome, Italy

AT NEXT INGEGNERIA DEI SISTEMI — *Oct. 2024 - Jan. 2026*

- Worked for **Next Ingegneria dei Sistemi** as a consultant for **MBDA Italy** in the embedded software department.
- Developed software components written mainly in `C++`, `Java` and `Ada` and then designed, performed and automatized corresponding unit test procedures.
- Participated in system integration activities, interfacing and validating multiple software components.

Technologies: C++, Java, Ada, Qt Creator, Apache NetBeans, Gnat Studio, SQL, Bash, XML, IBM RTC, IMB Doors, Wireshark, Linux, Windows, Microsoft Office

Ph.D. Researcher

Milan, Italy

IN THEORETICAL PARTICLE PHYSICS AT THE UNIVERSITY OF MILAN AND INFN — *Oct. 2021 - Sept. 2024*

- Worked under the supervision of Prof. Stefano Forte in the **NNPDF** collaboration as a developer of the `NNPDF` code.
- Developed techniques and computational programs that utilize artificial intelligence to investigate the internal structure of the proton analysing experimental data collected at **CERN**.
- Developed programs for solving the so-called DGLAP equations, a linear system of integro-differential equations, with numerical techniques.
- Published research results in various papers and presented them at international conferences.

Technologies: Python, Numpy, Scipy, Matplotlib, Keras, Tensorflow, Numba, Fortran, Bash, Git, Github, GitHub Actions, YAML, JSON, Slurm, PBS, Wolfram Mathematica, Linux, MacOS, VS Code, Vim, Latex, SQL, SQLite

Undergraduate Researcher

Rome, Italy

IN THEORETICAL PARTICLE PHYSICS AT THE UNIVERSITY OF ROME "LA SAPIENZA" — *Mar. 2021 - Oct. 2021*

- Worked under the supervision of Dr. Marco Bonvini to produce high-precision theoretical predictions in particle physics.
- Focused on describing experimental data of electron-proton collisions, collected at the particle accelerators **HERA** and **SLAC**.
- Wrote from zero the `C++` library `Adani`, with the `Python` bindings available in the PyPI and in conda-forge, resulting in a published paper and presentations at international conferences.

Technologies: C++, GSL, Wolfram Mathematica, Linux, Bash, CMake, Emacs, Latex

Skills

Programming	C, C++, Python, Robot Framework, Ada, Bash, CMake, Docker
Operating systems	Linux, MacOS, Windows
Code editors	VS Code, Qt Creator, Gnat Studio, Emacs, Vim, Nano
Version control sysytems	Git, Github, Gitlab, Github Actions, Gitlab CI/CD, IBM RTC
Python packages	Numpy, Scipy, Matplotlib, Multiprocessing, Numba, Pandas, Keras, Tensorflow
Scientific programs	Matlab, Wolfram Mathematica
Writing	Latex, Markdown, Microsoft Office
Languages	Italian (native language), English (fluent)

Education

Ph.D. in Physics

Milan, Italy

University of Milan — *Oct. 2021 - Nov. 2024*

- Field of study: Theoretical Particle Physics, Computational Physics.
- Thesis: *Advancements in PDFs determination: Incorporation of QED effects and new theoretical improvements in a modern deep learning fitting framework*. **link**

M.S. in Physics

Rome, Italy

University of Rome "La Sapienza" — *Sep. 2019 - Oct. 2021*

- Field of study: Theoretical Particle Physics.
- Grade: 110/110 (cum laude).
- Thesis: *Construction of a next-to-next-to-next-to-leading order approximation for heavy flavour production in deep inelastic scattering with quark masses*. **Inspire**

B.S. in Physics

Rome, Italy

University of Rome "La Sapienza" — *Sep. 2016 - Nov. 2019*

- Grade: 110/110 (cum laude).
- Thesis: *Particle identification with the time of flight method and applications to the CMS experiment.*

Publications

2026	**Implementation of DIS at N^{3}LO for PDF determination**, A. Barontini, M. Bonvini, N. Laurenti, *Accepted in Eur. Phys. J. C*	***Inspire***
2025	**NNPDF progress and the path to proton structure at N^{3}LO accuracy**, A. Barontini, N. Laurenti, J. Rojo, Published in *PoS DIS2024*	***Inspire***
2024	**LO, NLO, and NNLO Parton Distributions for LHC Event Generators**, J. Cruz-Martinez, S. Forte, N. Laurenti, T. R. Rabemananjara, J. Rojo, Published in *JHEP*	***Inspire***
2024	**NNPDF4.0 aN^{3}LO PDFs with QED corrections**, A. Barontini, N. Laurenti, J. Rojo, Contribution to *DIS2024*	***Inspire***
2024	**The Path to N^{3}LO Parton Distributions**, The NNPDF Collaboration, R. D. Ball et al., Published in *Eur. Phys. J. C*	***Inspire***
2024	**Determination of the theory uncertainties from missing higher orders on NNLO parton distributions with percent accuracy**, The NNPDF Collaboration, R. D. Ball et al., Published in *Eur. Phys. J. C*	***Inspire***
2024	**Photons in the proton: implications for the LHC**, The NNPDF Collaboration, R. D. Ball et al., Published in *Eur. Phys. J. C*	***Inspire***
2023	**Inclusion of QED corrections in PDFs fits**, N. Laurenti, Published in *Nucl. Part. Phys. Proc.*	***Inspire***
2022	**Approximating missing higher-orders in transverse momentum distributions using resummations**, N. Laurenti, T. R. Rabemananjara, and R. Stegeman, Contribution to *DIS2022*	***Inspire***

Teaching activity

2024	**Co-supervisor of a Bachelor thesis**, Thesis title: *On the fitting scale dependence of the Parton Distributions*	*University of Milan*
2024	**TA for the course of Quantum Physics I**, Introduction to Quantum Mechanics	*University of Milan*
2024	**TA for the course of Physics**, Basics of Classical Mechanics and Thermodynamics	*University of Milan*
2024	**TA for the course of Quantum Physics II**, Advanced course on Quantum Mechanics	*University of Milan*
2023	**TA for the course of Theoretical Physics I**, Introduction to Quantum Field Theory	*University of Milan*
2023	**TA for the course of Physics**, Basics of Classical Mechanics and Thermodynamics	*University of Milan*
2023	**TA for the course of Quantum Physics II**, Advanced course on Quantum Mechanics	*University of Milan*
2023	**Exercise classes for the course of Quantum Physics II**, Advanced course on Quantum Mechanics	*University of Milan*
2022	**TA for the course of Quantum Physics I**, Introduction to Quantum Mechanics	*University of Milan*
2021	**Student collaboration scholarship**, TA for the Laboratory courses of the first three years	*University of Rome "La Sapienza"*